

HIGH YIELD

SUPER HIGH YIELD

Localize the lesion before naming the disease.

NEPHROTIC

- Proteinuria >3.5 g/day
- Hypoalbuminemia -> edema
- Hyperlipidemia + lipiduria
- Podocyte / filtration-barrier injury
- Fatty casts / oval fat bodies -> Maltese cross
- Hypercoagulability from urinary anticoagulant loss

NEPHRITIC

- Hematuria + dysmorphic RBCs + RBC casts
- Proteinuria usually sub-nephrotic
- Azotemia + hypertension
- Inflammatory glomerular injury
- RBC cast = glomerular bleeding

TUBULOINTERSTITIAL

- WBC casts -> pyelonephritis / interstitial nephritis
- Muddy brown casts -> ATI/ATN
- Concentrating defects -> polyuria/nocturia
- Usually mild proteinuria
- Tubules + interstitium, not primary glomerulus

VASCULAR / POSTRENAL

- HTN + ischemia -> nephrosclerosis / RAS
- MAHA + thrombocytopenia -> TMA
- Obstruction -> hydronephrosis
- Sudden obstruction -> renal colic
- Unrelieved obstruction -> renal atrophy

DEPOSIT LOCATION + IF PATTERN

- BLOOD -> endothelium -> GBM -> podocyte -> URINE
- SUBENDOTHELIAL = blood-side -> inflammation -> nephritic tendency
- SUBEPITHELIAL = podocyte-side -> protein leak -> nephrotic tendency
- Granular IF = immune-complex pattern
- Linear IF = anti-GBM pattern
- Mesangial deposits: IgA nephropathy
- PSGN exception: subepithelial humps but nephritic
- MPGN I: subendothelial; dense deposit disease: intramembranous

URINE MICROSCOPY + AKI LOCALIZATION

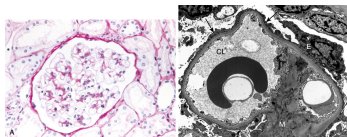
- RBC casts -> nephritic / glomerulonephritis
- WBC casts -> pyelonephritis or acute interstitial nephritis
- Muddy brown granular casts -> acute tubular injury/necrosis
- Fatty casts / oval fat bodies -> nephrotic syndrome
- Prerenal AKI: inadequate renal blood flow; BUN:Cr often >20:1
- Intrinsic renal AKI: renal parenchymal injury; BUN:Cr often ~10-15:1
- Postrenal AKI: urinary obstruction; ratio variable

CORE: heavy proteinuria -> hypoalbuminemia -> edema + hyperlipidemia/lipiduria + thrombotic/infection risk

Minimal Change Disease

- Child; common nephrotic syndrome in children
- LM near-normal; IF no pattern
- EM diffuse foot-process effacement

LM + EM

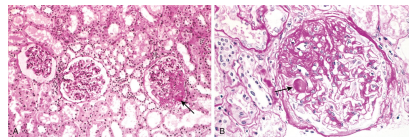


LOOK FOR: near-normal LM + podocyte foot-process effacement

FSGS

- Focal + segmental sclerosis
- Secondary associations: HIV, obesity, reflux, heroin
- Often progresses to CKD; poorer steroid response

LM

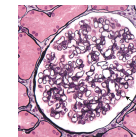


LOOK FOR: segmental scar/collapse involving only part of some glomeruli

Membranous Nephropathy

- Diffuse GBM thickening; primary often anti-PLA2R
- Granular IgG + C3; subepithelial deposits
- Classic SPIKE & DOME

LM / silver stain

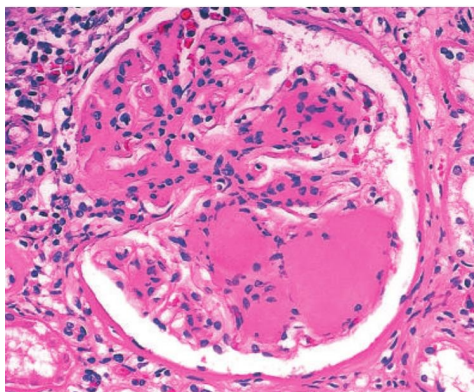


LOOK FOR: GBM spikes between subepithelial immune deposits

Diabetic Nephropathy

- Hyperglycemia -> AGEs + TGF-beta -> matrix accumulation
- Earliest structural change: GBM thickening
- Kimmelstiel-Wilson = nodular mesangial sclerosis
- Also diffuse mesangial sclerosis + hyaline arteriosclerosis

LM / PAS

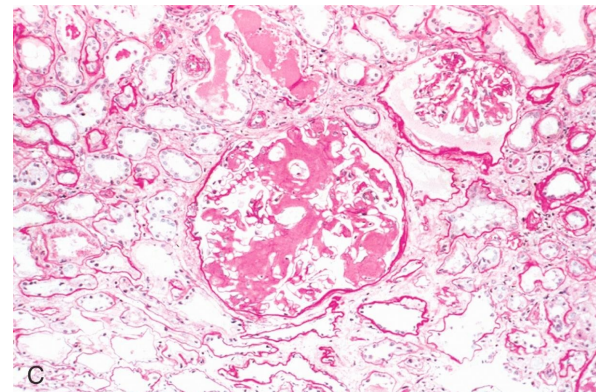


LOOK FOR: round Kimmelstiel-Wilson mesangial nodule

Renal Amyloidosis

- Amyloid deposits in glomeruli/mesangium + basement membranes
- Proteinuria is the common renal presentation
- Advanced deposits can distort/obliterate glomerular architecture

LM



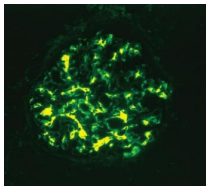
LOOK FOR: amorphous extracellular deposits expanding the glomerulus

CORE: hematuria + RBC casts; proteinuria usually sub-nephrotic; azotemia + hypertension

IgA Nephropathy (Berger)

- Synpharyngitic / recurrent hematuria
- LM mesangioproliferative
- IF granular IgA + C3 in mesangium

IF

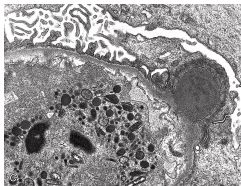


LOOK FOR: granular mesangial IgA fluorescence

Postinfectious / PSGN

- 1-4 weeks after nephritogenic Group A strep
- Low serum complement
- Granular IgG + C3; subepithelial humps

EM

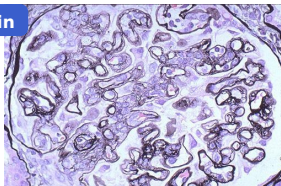


LOOK FOR: large subepithelial electron-dense HUMPS

MPGN

- Mixed nephritic/nephrotic presentation
- TRAM-TRACK / double-contour GBM
- Type I subendothelial; dense deposit disease intramembranous

LM / silver stain



LOOK FOR: duplicated GBM / tram-track contour

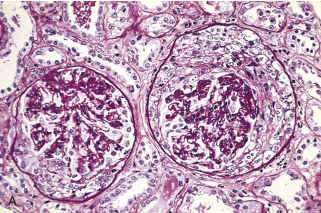
FAST DIFFERENTIATION

Disease	Timing / setting	IF	Deposit / buzzword
IgA	Same-time / synpharyngitic hematuria	Mesangial IgA + C3	Mesangial
PSGN	Delayed 1-4 wk after strep	Granular IgG + C3	Subepithelial humps
MPGN I	Chronic immune-complex setting	C3 + IgG/C1q/C4	Subendothelial
Dense deposit	Alternative complement dysregulation	C3 dominant	Intramembranous

Rapidly Progressive GN (RPGN)

- Rapid renal failure + severe oliguria + nephritic syndrome
- LM CRESCENTS - cellular early, fibrous later
- Crescents contain proliferating parietal cells + macrophages + fibrin

LM



LOOK FOR: crescent filling/compressing Bowman space

TYPE I - anti-GBM

- LINEAR IF; Goodpasture when lung + kidney
- Anti-GBM antibody injury

TYPE II - immune complex

- GRANULAR IF
- Postinfectious GN, IgA/HSP, lupus

TYPE III - pauci-immune

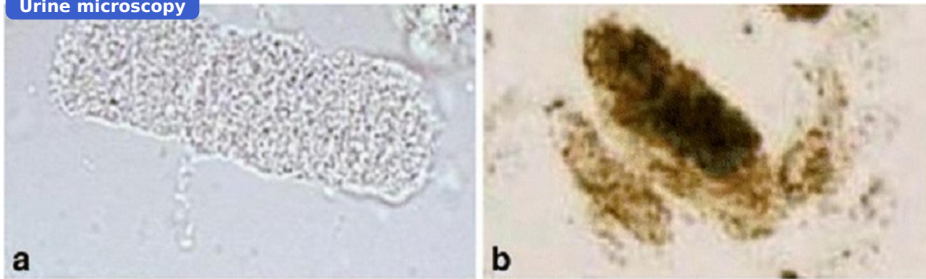
- Little/no IF staining; ANCA-associated
- GPA, microscopic polyangiitis, eosinophilic GPA

LUPUS NEPHRITIS - 6 CLASSES FULL HOUSE: IgG, IgM, IgA, C3, C1q, C4		
Class	Pattern	High-yield clue
I	Minimal mesangial	Normal LM
II	Mesangial proliferative	Mesangial hypercellularity
III	Focal nephritis	<50% glomeruli; nephritic
IV	Diffuse nephritis	>=50%; severe nephritic / wire loops
V	Membranous nephritis	Nephrotic pattern
VI	Sclerotic nephritis	Global sclerosis / ESRD

Acute Tubular Injury / Necrosis

- Ischemic or nephrotoxic tubular epithelial injury
- **MUDDY BROWN granular / epithelial casts**
- Tubular injury + obstruction/back-leak -> reduced GFR -> oliguria

Urine microscopy

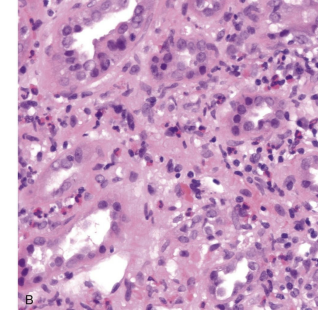


LOOK FOR: granular brown cast formed from tubular-cell debris

Drug-Induced AIN

- Usually follows drug exposure
- **Fever + rash + eosinophilia**
- WBC casts / leukocyturia support interstitial inflammation

LM / H&E

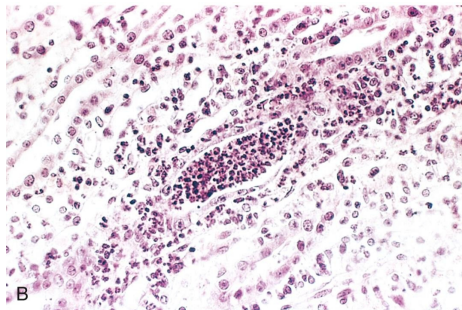


LOOK FOR: inflammatory cells expanding the interstitium around tubules

Acute Pyelonephritis

- Suppurative kidney infection; usually **ASCENDING**
- **Fever + flank/CVA pain + pyuria / WBC casts**
- Histology: neutrophils in tubules + interstitium

LM / H&E

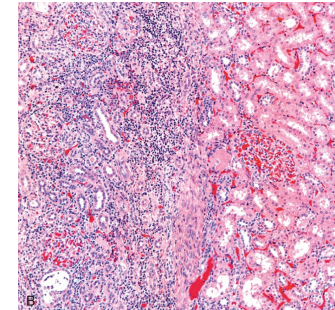


LOOK FOR: neutrophilic infiltrate extending through tubules/interstitium

Chronic Pyelonephritis

- Recurrent infection with calyceal/interstitial scarring
- Asymmetrically contracted kidneys + calyceal scars
- **Tubular atrophy + interstitial fibrosis + THYROIDIZATION**

LM / trichrome/H&E

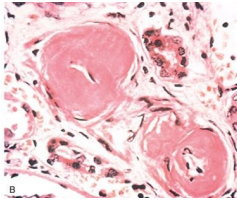


LOOK FOR: scarred cortex with atrophic tubules and chronic inflammation

Benign Nephrosclerosis

- Chronic HTN -> narrowed arterioles -> chronic ischemia
- **Hyaline arteriolosclerosis**
- Gross kidney: fine, even cortical granularity

LM / H&E

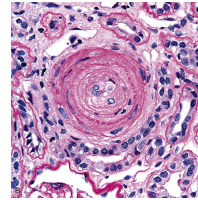


LOOK FOR: homogeneous pink hyaline thickening narrowing the arteriole

Malignant Nephrosclerosis

- Rapid severe HTN with end-organ injury
- **ONION-SKIN** hyperplastic arteriolosclerosis
- Can also show fibrinoid necrosis

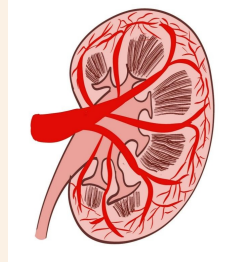
LM / H&E



LOOK FOR: concentric laminated "onion-skin" arteriolar wall

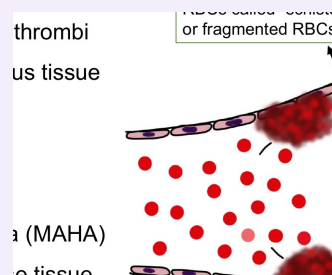
Renal Artery Stenosis

- **Ischemic kidney -> RENIN -> secondary hypertension**
- >70% atherosclerosis; fibromuscular dysplasia in younger women
- Stenotic kidney: diffuse ischemic atrophy
- **Contralateral kidney: hypertensive vascular injury**



THROMBOTIC MICROANGIOPATHY (TMA)

- **TMA TRIAD: microvascular thrombosis + MAHA + thrombocytopenia**
- Schistocytes = mechanically fragmented RBCs
- Microvascular occlusion -> tissue ischemia; platelets are consumed



RBCs shear across platelet-rich microthrombi -> schistocytes

HUS

- **Shiga-toxin HUS** after bloody diarrhea from EHEC O157:H7
- **RENAL** failure more severe; **CNS** less pronounced
- Atypical HUS = complement-mediated TMA

TTP

- **ADAMTS13** deficiency
- **CNS** more pronounced; renal failure less severe
- Lecture associations: SLE, HIV, hematologic malignancy

ADPKD

- Autosomal dominant; adult presentation; bilateral expanding cysts
- Polycystin proteins; progressive destruction of renal parenchyma
- Extrarenal: hepatic cysts + BERRY aneurysms
- Also pancreatic/splenic/lung cysts; colonic diverticula



ARPKD

- Autosomal recessive; neonates / infants / children
- PKHD1 -> fibrocystin
- Enlarged smooth kidneys with dilated elongated channels
- Congenital hepatic fibrosis is a key extrarenal feature

Other Cystic Lesions

- Benign simple cysts: older age; common post-mortem finding
- Medullary sponge kidney
- VHL-associated cystic disease
- Tuberous sclerosis-associated cystic disease (TSC1/TSC2)

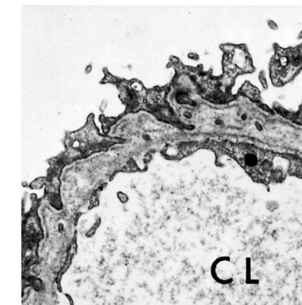
Alport Syndrome

- Type IV collagen basement-membrane disorder; usually X-linked
- Hematuria + deafness + ocular abnormalities
- EM BASKET-WEAVE GBM

EM

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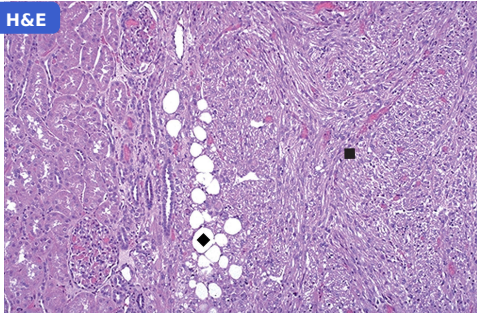


LOOK FOR: irregular GBM splitting + lamellation = basket weave

Angiomyolipoma

- Benign adult renal tumor
- **ANGIO + MYO + LIPO = vessels + smooth muscle + fat**
- Perivascular epithelioid-cell origin

LM / H&E

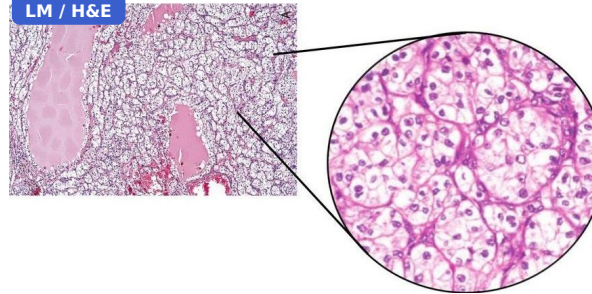


LOOK FOR: intermixed thick-walled vessels, smooth muscle, and adipose

Clear Cell RCC

- Proximal tubule; VHL loss -> HIF accumulation
- Golden-yellow gross tumor; clear cells in nests
- May extend into renal vein -> IVC

LM / H&E

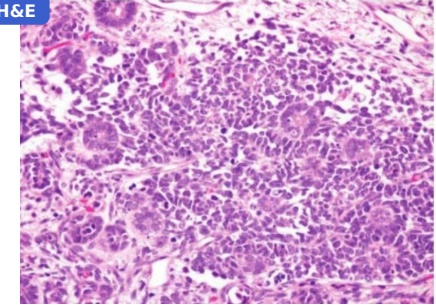


LOOK FOR: nests of clear cells separated by delicate vascular network

Wilms Tumor

- Child <6 years; WT1 / 11p13 association
- **TRIPHASIC tumor**
- TP53-associated anaplasia -> chemoresistance

LM / H&E



LOOK FOR: blastemal + epithelial + stromal components

Other Adult Renal Tumors

- Papillary RCC: proximal tubule
- Chromophobe RCC: intercalated cell
- Oncocytoma: benign; intercalated cell
- Adult malignant renal tumor category: RCC



Urothelial Carcinoma

- Painless hematuria; smoking + aryl-amine exposure
- Muscularis propria invasion = major staging line
- Flat CIS can form no mass -> normal CT

Cystoscopy + LM



LOOK FOR: papillary fronds built around fibrovascular cores

Bladder Inflammation / SCC

- Acute cystitis: culture positive, usually E. coli; neutrophils + hyperemia
- Interstitial cystitis: >6 wk, sterile cultures, fissures/punctate hemorrhages, mast cells
- **Exclude flat urothelial CIS before diagnosing interstitial cystitis**
- Bladder SCC precursor in lecture: squamous metaplasia



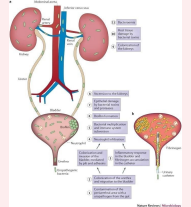
Cystitis - Lower UTI

- Bladder mucosal inflammation; *E. coli* is #1 infectious cause
- **Dysuria + frequency + urgency + suprapubic tenderness**
- Lower UTI = bladder and/or urethra
- Acute hemorrhagic cystitis can present with gross hematuria



Pyelonephritis - Upper UTI

- Kidney/renal interstitium infection; usually **ASCENDING**
- **Fever/chills + flank pain + CVA tenderness**
- May retain lower-UTI symptoms
- **WBC / leukocyte casts point to renal involvement**
- Complications: urosepsis, abscess, papillary necrosis, recurrent pyelo



UPEC / *E. coli*

- **#1 cause of UTI**
- Gram-negative; lactose fermenter; nitrite positive
- Type 1 fimbriae; P fimbriae linked to pyelonephritis
- Hemolysin + siderophores



S. saprophyticus

- Young sexually active women; honeymoon cystitis
- Gram-positive cocci; catalase +; coagulase -
- **Novobiocin resistant; urease +; nitrite -**

Proteus mirabilis

- **Urease positive + SWARMING motility**
- Gram-negative rod; non-lactose fermenter
- **Urea splitting -> STRUVITE stones**



UA / Other agents

- Leukocyte esterase = WBC enzyme
- Nitrites = nitrate-reducing bacteria
- Lecture agents: *Klebsiella*, *Enterococcus*, *Pseudomonas*
- Microscopy: pyuria, hematuria, leukocyte casts

NEPHROTIC MCD -> child + normal LM + foot-process effacement

NEPHROTIC Membranous -> spike & dome + granular IgG/C3 + subepithelial

NEPHROTIC Amyloid -> mesangial/BM deposits + proteinuria

NEPHRITIC PSGN -> 1-4 wk after strep + low complement + humps

RPGN Crescents: I linear, II granular, III pauci-immune

TUBULAR ATI/ATN -> muddy brown granular casts

PYELO Acute -> CVA tenderness + WBC casts + neutrophils

VASCULAR Benign HTN -> hyaline arteriolosclerosis + fine granular kidney

RAS Ischemic small kidney + renin HTN; contralateral hypertensive injury

HUS/TTP HUS kidney > CNS; TTP ADAMTS13 + CNS > kidney

TUMOR Clear cell RCC -> VHL/HIF + golden-yellow + renal vein invasion

BLADDER Painless hematuria; muscularis propria invasion = staging line

UTI Cystitis suprapubic; pyelo flank/CVA + WBC casts

NEPHROTIC FSGS -> focal + segmental sclerosis; HIV/obesity/heroin/reflux

NEPHROTIC Diabetes -> Kimmelstiel-Wilson + GBM thickening + hyaline arteriosclerosis

NEPHRITIC IgA -> synpharyngitic hematuria + mesangial IgA/C3

NEPHRITIC MPGN -> tram-track; type I subendothelial / dense deposit intramembranous

LUPUS Full house; III/IV nephritic; V nephrotic

INTERSTITIAL AIN -> fever + rash + eosinophilia + WBC casts

PYELO Chronic -> asymmetric scars + calyx deformity + thyroidization

VASCULAR Malignant HTN -> onion-skin + fibrinoid necrosis

TMA MAHA + thrombocytopenia + microthrombi; schistocytes

CYSTIC ADPKD -> hepatic cysts + berry aneurysm; Alport -> basket weave

TUMOR Wilms -> triphasic + WT1; TP53 anaplasia

BLADDER Flat CIS can have normal imaging

FAST ORDER: syndrome -> urine cast -> complement -> IF pattern -> deposit location -> buzzword